10	A body is moving in a circular path with a radius of 7.70 cm. The time for one orbit is 2.36 s. What is the centripetal acceleration of the body?							
	A	0.109 m s^{-2}	**B**	0.206 m s^{-2}	**C**	0.368 m s^{-2}	**D**	0.546 m s^{-2}

L2	**Answer: D** **centripetal acceleration** $a = r\omega^2$ $= r(2\pi/T)^2$ $= (7.70 \times 10^{-2})(2\pi/2.36)^2$ $= 0.546\ m\ s^{-2}$